Industrial Internship Report on Predictive Maintenance of Gearbox using Machine Learning

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Internship Organization: upskill Campus & The IoT Academy (in collaboration with UniConverge Technologies Pvt Ltd)

Domain: Machine Learning / Artificial Intelligence

Duration: 6 Weeks

Executive Summary

This report provides details of the 6-week Industrial Internship provided by upskill Campus and The IoT Academy in association with UniConverge Technologies Pvt Ltd (UCT). My project focused on developing a Machine Learning solution for the Predictive Maintenance of Gearboxes using vibration sensor data. The project aims to classify gearbox conditions as Healthy or Faulty using Python, Pandas, NumPy, and Scikit-learn. Logistic Regression and Random Forest models were utilized for classification, and a Streamlit dashboard was designed to display real-time predictions. The goal is to help identify potential gearbox faults to reduce industrial downtime.

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1. Preface

During this 6-week program, I worked step-by-step from requirements analysis to final ML model deployment:

- **Phase 1:** Understanding project deliverables and exploring UCT's predictive maintenance use cases.
- **Phase 2:** Data collection, cleaning, and preprocessing using Pandas and NumPy.
- **Phase 3:** Feature engineering based on vibration (X, Y, Z axis) and temperature data.
- **Phase 4:** Training Logistic Regression and Random Forest models using Scikit-learn.
- **Phase 5:** Model evaluation and building a frontend dashboard using Streamlit.
- **Phase 6:** Project packaging, GitHub repository documentation, and final report submission.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd (UCT)

Established in 2013, UniConverge Technologies specializes in Digital Transformation, Industrial IoT (IIoT), Machine Learning, and Cloud Computing. UCT develops enterprise platforms such as Factory Watch for smart asset monitoring and predictive maintenance.

2.2 About upskill Campus & The IoT Academy

upskill Campus (USC) is a career development platform delivering scalable training and industry internships. In partnership with The IoT Academy, USC provides job-oriented technical programs to bridge academia and industry demands.

2.3 Objectives

- To gain practical experience in building industry-grade Machine Learning models.
- To solve real-world industrial problems like machine failure prediction.
- To deploy Python-based ML models into interactive web applications (Streamlit).

2.4 Glossary

- **ML:** Machine Learning
- **RF:** Random Forest Classifier
- **CSV:** Comma-Separated Values

3. Problem Statement

Unexpected gearbox failures in industrial machines cause significant downtime and financial loss. The requirement was to analyze sensor data (vibration and temperature) to predict whether the gearbox is "Healthy" or "Faulty" before a critical failure occurs.

4. Existing and Proposed Solution

- **Existing Systems:** Manual inspection with preventive maintenance (replacing parts on a schedule), which is costly and reactive.
- **Proposed Solution:** A condition monitoring and predictive maintenance system using Machine Learning.
- **Value Addition:** Using a Random Forest Classifier to predict faults with high accuracy and a Streamlit dashboard for operators to visualize the machine's health status instantly.
- **Code Repository:** https://github.com/karthikavengatesan2008/internship-project
- **Report Repository:** https://github.com/karthikavengatesan2008/internship-project

5. Proposed Design & Architecture

5.1 High-Level Architecture (HLD)

```
      1. Input: Vibration and Temperature Sensor Data (CSV) | ▼ 2. Processing:
Data Preprocessing & Feature Extraction (Pandas/NumPy) | ▼ 3. Modeling: Scikit-
Learn (Random Forest Classifier) | ▼ 4. Output: Streamlit Web Dashboard
displaying Healthy/Faulty status
```

5.2 Low-Level Component Flow

The system ingests continuous sensor readings. The data is normalized to handle discrepancies in scale between temperature and vibration metrics. The pre-trained Random Forest model then evaluates these features against learned failure patterns, outputting a binary classification which is immediately reflected on the user interface.

6. Performance & Quality Testing

6.1 Test Cases & Plan

Test ID	Test Scenario	Expected Result	Status
TC01	Input normal vibration data	Model predicts "Healthy"	Passed
TC02	Input high vibration/temp data	Model predicts "Faulty"	Passed
TC03	Dashboard UI interaction	Streamlit updates prediction dynamically	Passed

6.2 Performance Outcome

- The Random Forest model achieved high accuracy and robust precision in classifying the sensor data compared to a baseline Logistic Regression.
- The Streamlit dashboard successfully processed user inputs and returned real-time ML predictions without noticeable latency.

7. My Learnings

- Gained hands-on experience in data preprocessing and model training using Python and Scikit-learn.
- Learned how to evaluate ML models using accuracy and performance metrics.
- Acquired skills to build and deploy interactive data apps using Streamlit.
- Developed professional version control habits by organizing projects on GitHub.

8. Future Scope of Work

- Implementing Deep Learning models (like LSTM) for time-series forecasting of sensor data over longer periods.
- Deploying the Streamlit application to cloud platforms like AWS or Heroku for remote operator access.
- Integrating real-time IoT sensor data streams directly into the ML prediction pipeline.